Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

---

1. Administrative & Study Identification

Full Study Title: HERTHENA-Breast-01: A Phase 1b/2, Multicenter, Open-label, Dose-Finding Study to Evaluate the Safety and Antitumor Activity of Patritumab Deruxtecan in Participants With HER2 Positive Unresectable Locally Advanced Breast Cancer or Metastatic Breast Cancer **Plain Study Title:** Study of Patritumab Deruxtecan With Other Anticancer Agents in Participants With HER2 Positive Breast Cancer That Has Spread and Cannot Be Surgically Removed (MK-1022-009) **Short Title/Acronym:** HERTHENA-Breast01 (MK-1022-009) **Protocol Number:** MK-1022-009 / 1022-009 / jRCT2041250022 **NCT Number:** NCT06686394 **Posting ID:** 805073040590599769 **Trial Status:** Recruiting **Sponsor Name:** Merck Sharp & Dohme LLC / Daiichi Sankyo (Company: Merck) **Investigational Product:** Patritumab Deruxtecan (HER3-DXd / MK-1022) in combination with other anti-HER2 agents (Trastuzumab, Pertuzumab, Tucatinib) **Phase of Development:** Phase 1, Phase 2 (Phase 1b/2) **Medical Monitor:** Clinical Operations Lead, Merck

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate dose-limiting toxicities, safety, and tolerability (including treatment discontinuations due to adverse events) of patritumab deruxtecan in combination with anti-HER2 agents. **Secondary Objectives:** To evaluate the pharmacokinetic exposure parameters of HER3-DXd ADC, total HER3-DXd antidrug antibody (ADA), and free DXd payload, as well as preliminary antitumor activity.

2.2 Background & Rationale Although HER2-targeted antibody-drug conjugates (ADCs) have improved outcomes in HER2+ advanced or metastatic breast cancer, safe and effective later-line therapies are needed for patients who have progressed on trastuzumab deruxtecan (T-DXd). Human epidermal growth factor receptor 3 (HER3) is expressed in many tumor types, including HER2+ breast cancer. Patritumab deruxtecan (HER3-DXd) is a novel ADC that selectively binds HER3 and is composed of a fully human anti-HER3 IgG1 antibody linked to a cytotoxic topoisomerase I inhibitor. This study aims to evaluate its safety and efficacy in combination with standard anti-HER2 targeted agents.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Dose-finding (Multiple combination experimental arms) **Blinding:** Open-label **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration Target N: 81 participants (up to 27 participants in each arm) **Number of Sites:** Multicenter (Including USA, Canada, United Kingdom, Israel, Japan, South Korea) **Estimated Study Start:** 26-Feb-2025 **Estimated Primary Completion:** 18-Apr-2030

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with histologically confirmed HER2+ (per most recent ASCO/CAP guidelines: IHC 3+, IHC 2+/ISH+, or IHC undetermined/ISH+) unresectable locally advanced or metastatic breast cancer. 2. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 to 1 within 7 days before the start of study intervention. 3. Adequate organ function. 4. Human immunodeficiency virus (HIV)-infected participants must have well-controlled HIV on antiretroviral therapy (ART). 5. Participants who are hepatitis B surface antigen (HBsAg) positive are eligible if they have received HBV antiviral therapy for at least 4 weeks and have an undetectable viral load. HCV patients are eligible if viral load is undetectable. *(Note: Arm 1 requires 2-5 prior lines including T-DXd progression; Arm 2 requires max 5 prior lines; Arm 3 requires max 3 prior lines including T-DXd as the most recent therapy).*

4.2 Exclusion Criteria 1. History of (noninfectious) pneumonitis/ interstitial lung disease (ILD) that required steroids or current pneumonitis/ILD. 2. Uncontrolled or significant cardiovascular disease or clinically severe respiratory compromise. 3. Any history of or evidence of current leptomeningeal disease, active brain metastases, or spinal cord compression. 4. Evidence of ongoing uncontrolled systemic bacterial, fungal, or viral infection. 5. Concurrent active HBV and HCV infection. 6. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 7. Clinically significant corneal disease. 8. Prior treatment with tucatinib, lapatinib, or neratinib, or investigational HER2-targeted TKIs (Applicable to Arm 3 only).

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patritumab Deruxtecan (HER3-DXd) administered intravenously, with dose finding based on the Bayesian Optimal Interval design. It is evaluated in 3 arms: * Arm 1: HER3-DXd + Trastuzumab (ATC Code: L01FD01; Trade names: Herceptin, Enhertu, Ontruzant, Trazimera, Herzuma, Phesgo, Kanjinti, Herceptin Hylecta, Ogivri, KADCYLA) * Arm 2: HER3-DXd + Pertuzumab + Trastuzumab * Arm 3: HER3-DXd + Tucatinib + Trastuzumab **Route of Administration:** Intravenous (IV) and Oral (Tucatinib) **Duration of Treatment:** Treatment will continue until radiographic progression, unacceptable toxicity, or participant withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Antiretroviral therapy for well-controlled HIV; HBV/HCV antiviral therapy. **Prohibited:** Concurrent investigational therapies; live vaccines; systemic therapy for active infections.

---

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy/Scans, Safety Labs
Baseline	Day 0	First Dose, Safety Baseline assessment
Treatment Cycles	Every 3 Weeks	Safety Labs, Efficacy Assessment, PK/ADA blood collection
End of Study	Follow-up	Radiographic progression evaluation, Treatment discontinuation, Survival follow-up

---

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Number of Participants Experiencing Dose-Limiting Toxicity (DLT) 2. Number of Participants with One or More Adverse Events (AEs) 3. Number of Participants who Discontinue Study Intervention Due to an AE **Measurement Method:** Clinical assessments with AEs graded per National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) version 5.0.

7.2 Secondary & Exploratory Endpoints * Pharmacokinetics (PK) of Patritumab Deruxtecan Antibody-Drug Conjugate (ADC), specifically: 1. Maximum Plasma Concentration (Cmax) of Patritumab Deruxtecan ADC 2. Trough Concentration (Ctrough) of Patritumab Deruxtecan ADC 3. Area Under the Plasma Concentration-Time Curve (AUC) of Patritumab Deruxtecan ADC * Total HER3-DXd antidrug antibody (ADA) and free DXd payload. * Preliminary antitumor activity (Objective Response Rate, Progression-Free Survival, etc.).

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring and grading per NCI CTCAE 5.0. Specifically, strict monitoring for pneumonitis/ILD. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for SAEs. **Data Monitoring Committee (DMC):** Internal/External Safety Review Committee responsible for making dose escalation and de-escalation decisions using the Bayesian Optimal Interval design.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all participants receiving ≥ 1 dose; Efficacy Evaluable Set for tumor response. **Sample Size Justification:** Up to 81 participants (up to 27 per arm) calculated to support the Bayesian Optimal Interval design for Phase 1b/2 dose-finding. **Handling of Missing Data:** Standard handling conventions for oncology PK/ADA and efficacy readouts, censoring patients at last evaluable follow-up for survival statistics.

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring incorporating onsite and remote data verification. **Audit Trail:** Robust electronic case report form (eCRF) locking and query resolution.

---

11. Study Identifiers & Governance

Primary ID: MK-1022-009 **Registry Number:** NCT06686394 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC / Daiichi Sankyo **Study Title:** HERTHENA-Breast-01 **Official Regulatory Title:** A Phase 1b/2, Multicenter, Open-label, Dose-Finding Study to Evaluate the Safety and Antitumor Activity of Patritumab Deruxtecan in Participants With HER2 Positive Unresectable Locally Advanced Breast Cancer or Metastatic Breast Cancer

---

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Breast Neoplasms / Breast Cancer (HER2-Positive Metastatic Breast Cancer) **Diagnosis Threshold:** IHC 3+, IHC 2+/ISH+, or IHC undetermined/ISH+ confirmed via recent ASCO/CAP guidelines **Medical Methodology:** HER3-directed Antibody-Drug Conjugate (ADC) plus Anti-HER2 Targeted Therapy **Optimization Target:** Recommended Phase 2 Dose (RP2D) / Maximum Tolerated Dose determination

---

13. Study Procedures & Methodology

Management Pathway: Sequential cohort dose-finding via Bayesian Optimal Interval design **Education Protocol (HCP):** Site-specific training on ADC safety profile, specifically identifying and managing ILD/pneumonitis **Education Protocol (Patient):** Comprehensive informed consent regarding ADC therapy and the reporting of pulmonary symptoms **Quality Audit Mechanism:**

Centralized RECIST 1.1 review for radiographic imaging and standard GCP audit procedures

---

14. Observation & Follow-Up Schedule

Total Duration: Follow-up estimated through April 2030 **Onsite Visit Cadence:** Corresponds with treatment cycles (typically Q3W/Q4W based on combination agent schedules) **Remote Monitoring Frequency:** As per clinical data monitoring plan **Checklist Review Interval:** AE review and concomitant medication tracking evaluated at the start of every cycle

---

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]